

SIMATS ENGINEERING

ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 Total Marks:30 Annexure A

Dry Run Evaluation

Code of Conduct:

*I SHAILAJA S (192511026)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph: A

/ \

B C

/ \ \

D E F G

Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration Queue Visited Node

- 1
- 2
- 3
- 4
- 5
- 6
- 7

Questions:

- 1. Write the BFS traversal order.
- 2. Show the queue contents after each iteration.

Iteration	Queue	Visited Node
1	A	A
2	B, C	B
3	C, D, E	C
4	D, E, F, G	D
5	E, F, G	E
6	F, G	F
7	G	G

1. BFS Traversal Order

A → B → C → D → E → F → G

2. Queue Contents After Each Iteration

- ❖ Iteration 1: A
- ❖ Iteration 2: B, C
- ❖ Iteration 3: C, D, E
- ❖ Iteration 4: D, E, F, G
- ❖ Iteration 5: E, F, G
- ❖ Iteration 6: F, G
- ❖ Iteration 7: G

- ✓ BFS visits nodes level by level
- ✓ Starts from root A, then explores B, C, followed by D, E, F, G

2. Depth-First Search (DFS) Using the same graph,

A

/ \

B C

/ \ / \

D E F G

Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

Step 1

Stack = [A]

Pop **A**, visit it

Stack = [C, B]

Step 2

Pop **B**, visit it

Visited: **A, B**

Push E, D

Stack = [C, E, D]

Step 3

Pop **D**, visit it

Visited: **A, B, D**

D has no children

Stack = [C, E]

Step 4

Pop **E**, visit it

Visited: **A, B, D, E**

E has no children

Stack = [C]

Step 5

Pop C, visit it

Visited: **A, B, D, E, C**

Push G, F

Stack = [G, F]

Step 6

Pop F, visit it

Visited: **A, B, D, E, C, F**

Stack = [G]

Step 7

Pop G, visit it

Visited: **A, B, D, E, C, F, G**

Stack = []

Iteration	Stack (Top → Right)	Visited Node
1	A	A
2	C, B	B
3	C, E, D	D
4	C, E	E
5	C	C
6	G, F	F
7	G	G

3. Initial State :

1 3 6
5 0 2
4 7 8

Goal State :

1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion

1
2
3
4
5

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

Initial State

1 3 6
5 0 2
4 7 8

Goal State

1 2 3
4 5 6
7 8 0

BFS State Expansion Table

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	Initial State	[Initial]	S1, S2, S3, S4
2	S1	[S1, S2, S3, S4]	S2, S3, S4, S5, S6
3	S2	[S2, S3, S4, S5, S6]	S3, S4, S5, S6, S7
4	S3	[S3, S4, S5, S6, S7]	S4, S5, S6, S7, S8
5	Goal Found	[...]	Goal State reached

1. Which node is expanded in each iteration?

- Iteration 1 → Initial State
- Iteration 2 → S1
- Iteration 3 → S2
- Iteration 4 → S3
- Iteration 5 → Goal State found

2. How many states are generated in total?

- Around **10–20 states** are generated (depends on duplicate removal).
- BFS generates **all possible states level by level** until goal is reached.

3. Complete Solution Path

```
Initial State
↓
Move 0 Left
↓
Move 0 Up
↓
Move 0 Right
↓
Goal State
```

Example path:

```
1 3 6      1 3 6      1 3 6      1 2 3
5 0 2 → 0 5 2 → 5 2 0 → 4 5 6
4 7 8      4 7 8      4 7 8      7 8 0
```

4. Why BFS gives shortest solution?

- BFS explores level by level
- First time reaching goal = minimum number of moves
- All moves have equal cost (unweighted problem)

RUBRICS FOR CALCULATION

Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)
20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.
30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.
20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.
20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.

Criteria Level 1 – Beginning (1)

Shows little or

**Understanding
of Search
Problem**

**Application of Search
Algorithm and Logical
Reasoning**

**State
Expansion and
Dry
Run
Execution**

**Accuracy of
Solution
Path**

10 Limited
Queue/Stack/Priority Queue explanation and organization; updates, state reasoning is diagrams, and partially clear. explanations are no neat, organized, understanding and logically of the search presented. problem.

Unable to
correctly
perform the dry
run or
expand states.

Fails to apply
the correct
search
algorithm.

Incorrect or
incomplete
solution path.

Poor
presentation
with unclear or
missing
reasoning.
Presentation is
clear with
minor
organizational
issues.